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Lab Manual: Module 7 — Electronics for Arduino Programming (Part 1)

Capacitors, Inductors, Diodes, LEDs & PWM

Introduction: Electronics as the Nervous System of Robotics

Before diving into individual components, it's worth understanding *why* this module matters. An Arduino microcontroller is essentially the "brain" of a robotic system — it makes decisions in code. But a brain without a nervous system cannot move a limb or feel the environment. Electronic components form that nervous system: they translate digital logic (HIGH/LOW signals) into physical actions (spinning motors, glowing LEDs, activated relays) and convert physical stimuli (light, temperature, distance) back into readable digital data.

A critical conceptual point: an Arduino pin set to HIGH is nothing more than a 5-volt electrical potential. It has no inherent understanding of motors, sensors, or safety limits. Passive components (resistors, capacitors, inductors) shape and stabilize this voltage, while active components (diodes, transistors, ICs) switch and route current safely. Without understanding this interaction, students risk practical hardware failures — burnt-out pins, destroyed LEDs, or damaged microcontrollers from voltage spikes. This module builds the theoretical foundation needed to avoid those failures.

Topic 1: Capacitors

1.1 Definition and Core Concept



A capacitor is a **passive** component that stores and releases electrical energy — but unlike a battery, it does not use chemical reactions. Instead, it stores energy **electrostatically**, within an electric field formed between two conductive plates separated by an insulating material called a **dielectric**. Conceptually, think of a capacitor as a tiny, fast-acting electrical reservoir: it fills up (charges) and empties (discharges) far more quickly than a chemical battery ever could, but it cannot hold nearly as much energy.

1.2 Capacitance and Units

The property that quantifies "how much charge a capacitor can store" is called **Capacitance (C)**, measured in **Farads (F)**. A full Farad is enormous for practical electronics, so real-world components are specified in fractional units:

- **Microfarads (μF)** = 10^{-6} F
- **Nanofarads (nF)** = 10^{-9} F
- **Picofarads (pF)** = 10^{-12} F

In Arduino-based circuits, you will typically encounter capacitors ranging from about 1 pF up to 1000 μF , depending on whether they're used for high-frequency filtering or power smoothing.

1.3 Circuit Symbols and Polarity

Schematically, capacitors are drawn as two parallel lines. There are two broad categories:

- **Non-polarized (e.g., ceramic)** — can be inserted in either orientation, since there's no positive/negative distinction.
- **Polarized (e.g., electrolytic)** — have a defined positive terminal (straight line) and negative terminal (curved line). Reversing polarity on these can cause the internal dielectric to break down, potentially resulting in an explosive failure. This is an important safety consideration for lab work.

1.4 Working Principle: Charging, Discharging, and AC/DC Behavior

The internal mechanics: when a voltage source is applied, electrons accumulate on the negative plate, which repels electrons away from the positive plate. This charge separation stores energy in the electric field between the plates.



In DC circuits:

- While charging, a capacitor initially behaves like a **short circuit** (current flows freely).
- Once fully charged, it behaves like an **open circuit**, completely blocking further DC current flow.
- This charging follows an exponential curve until the capacitor's voltage matches the supply voltage.

In AC circuits:

- Because AC voltage continuously reverses polarity, the capacitor is perpetually charging and discharging.
- This gives the *appearance* that AC signals pass through the capacitor, which is why capacitors are widely used to filter out high-frequency noise while blocking steady DC.

This dual behavior — blocking DC, passing AC — is the single most important conceptual property of capacitors and underlies nearly all their practical applications.

1.5 Types and Practical Applications

- **Ceramic capacitors:** Compact, non-polarized, commonly used for high-frequency noise filtering on motor terminals or microcontroller power pins.
- **Electrolytic capacitors:** Polarized, offer much higher capacitance, used for power supply smoothing and preventing "reset loops" when a circuit experiences a heavy current draw.
- **Tantalum capacitors:** High capacitance in a small footprint — useful in compact robotics designs.
- **Film capacitors:** Provide precise capacitance values, ideal for analog timing circuits.

Common lab applications: noise filtering on DC motors, smoothing voltage ripple in power supply circuits, stabilizing Arduino reset circuits, and hardware-based debouncing of tactile pushbuttons (using a capacitor to smooth out the rapid electrical "bounce" that occurs when a mechanical switch is pressed).

Topic 2: Inductors



2.1 Definition and Core Concept

An inductor is a passive component that stores energy in a **magnetic field** when current flows through it — the electromagnetic counterpart to the capacitor's electrostatic storage. Physically, it's built from insulated wire wound into a coil, often wrapped around a magnetic core (iron or ferrite) to concentrate and strengthen the magnetic field.

2.2 Inductance and Units

Inductance is measured in **Henries (H)**, named after physicist Joseph Henry. It represents a component's capacity to store magnetic energy and, critically, its **resistance to changes in current** (as opposed to a capacitor, which resists changes in voltage). Adding an iron or ferrite core boosts magnetic permeability, significantly increasing the inductance for the same coil geometry.

2.3 Circuit Symbols

- **Air-core inductor:** drawn as a series of plain loops.
- **Iron-core inductor:** drawn as loops with straight parallel lines above them, representing the core material.

2.4 Working Principle: Electromagnetic Induction

When current flows through the coil, it generates a magnetic field. If that current changes, the changing magnetic field induces a voltage (an electromotive force, or EMF) that **opposes the change** — this is Lenz's Law in action.

Conceptual analogy: Think of an inductor as a heavy flywheel inserted into a water pipe. When you first try to start the water flow, the flywheel's inertia resists it. But once the flywheel is spinning (current flowing), it wants to *keep* the water moving — even if you suddenly close the valve, the flywheel's momentum generates a pressure spike. This is exactly why inductors (like motor coils) generate dangerous voltage spikes when suddenly switched off — a concept that becomes critical later when discussing flyback diodes.

Key behavioral summary:

- Inductors allow **steady DC to pass freely** with minimal resistance.



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- Inductors **strongly oppose sudden surges or drops** in current.

2.5 Practical Applications

- **Motor windings and relay coils:** inherently inductive loads that generate magnetic fields to produce motion.
- **Switching power supplies (buck/boost converters):** inductors store and release energy thousands of times per second to step voltage up or down efficiently.
- **Noise filters / RF chokes:** combined with capacitors to form low-pass filters that block high-frequency electromagnetic interference (EMI) from motors, protecting the Arduino's logic lines.

Topic 3: Capacitors vs. Inductors — Comparative Analysis

This comparison is a key conceptual checkpoint before moving to semiconductors:

Property	Capacitor	Inductor
Energy storage	Electric field (electrostatic)	Magnetic field (electromagnetic)
Resists changes in	Voltage	Current
DC response	Blocks DC once charged	Passes DC freely



AC response	Passes AC	Blocks high-frequency AC
Unit	Farad (F)	Henry (H)
Symbol	Two parallel plates	Coiled wire loops
Typical use	Voltage smoothing, noise filtering	Current regulation, motor magnetics, switching converters

The core conceptual takeaway: capacitors and inductors are functional opposites. A capacitor "hates" sudden voltage change; an inductor "hates" sudden current change. This complementary relationship is why they're so often paired together in filter circuits.

Topic 4: Diodes

4.1 Definition and Semiconductor Basics

A diode is best understood as an **electrical one-way check valve** — it allows current to flow easily in one direction while blocking flow in the reverse direction. This is achieved through semiconductor physics: a diode is constructed by joining **P-type material** (rich in positive "holes") with **N-type material** (rich in negative electrons).

4.2 The PN Junction

Where the P-type and N-type materials meet, a **depletion region** forms — a barrier zone that naturally restricts electron flow until a specific voltage threshold is applied. Understanding this junction is the conceptual key to understanding all diode behavior.



4.3 One-Way Flow and Schematic Symbol

Conventional current flows only from the **Anode** (positive terminal) to the **Cathode** (negative terminal). The schematic symbol — a triangle pointing toward a vertical line — visually encodes this: the triangle represents the permitted direction of flow, and the flat line represents the block.

4.4 Forward Bias vs. Reverse Bias

Forward Bias (Conducting state): When the voltage source's positive terminal connects to the Anode and negative to the Cathode, the depletion region collapses, and current flows. Silicon diodes require a minimum of approximately **0.7V** to conduct, and once conducting, they maintain a fairly constant 0.7V drop across the junction.

Reverse Bias (Blocking state): When polarity is reversed, the depletion region widens, creating high internal resistance that blocks current almost entirely. The diode behaves like an open switch. However, if the reverse voltage becomes excessive, it exceeds the **breakdown voltage**, causing the junction to fail and allowing a destructive surge of reverse current — a condition that must always be avoided in circuit design.

4.5 Practical Applications — Diodes as Circuit "Bodyguards"

This is one of the most practically important sections for robotics work:

- **Flyback diode (motor protection):** When an inductive load like a motor or relay coil is switched off, it generates a dangerous voltage spike (recall the flywheel analogy from Topic 2). Placing a reverse-biased diode in parallel with the motor gives that spike a safe path to dissipate, protecting the Arduino from damage.
- **Reverse polarity protection:** A series diode at the power input blocks current entirely if a battery is connected backward, preventing a fatal short circuit.
- **ESD clamping:** Diodes on I/O pins shunt transient electrostatic spikes safely to VCC or GND before they can reach the sensitive silicon inside the microcontroller.

Topic 5: LEDs (Light Emitting Diodes)



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5.1 Working Principle: Electroluminescence

An LED is a specialized **active semiconductor diode** designed to emit visible light when forward-biased current flows through it. The underlying phenomenon is **electroluminescence**: when forward voltage is applied, electrons cross the PN junction and recombine with electron holes. This recombination releases energy — but instead of releasing that energy primarily as heat (as in a regular diode or incandescent filament), the LED releases it as light photons.

5.2 Efficiency and Advantages

LEDs consume up to 90% less energy than incandescent bulbs, generate very little heat, resist vibration well (no fragile filament), and have operational lifespans exceeding 50,000 hours — making them ideal for robotics applications where reliability and power efficiency matter.

5.3 Color Determination

An important conceptual correction to a common misconception: LED color is **not** determined by the tint of the plastic casing. It is determined by the physical semiconductor material used in the crystal (e.g., Gallium Phosphide for green, Indium Gallium Nitride for blue/white), which dictates the energy gap between electron states and therefore the specific photon wavelength emitted.

5.4 Polarity and Electrical Characteristics

LEDs are strongly polarized:

- **Anode (+)**: identified by the longer physical leg.
- **Cathode (-)**: identified by the shorter leg and a flat notch on the rim of the plastic lens.

Forward voltage (V_f) varies by color — this is a crucial calculation input:

- Red / Yellow: ~1.8V to 2.2V
- Green / Blue / White: ~3.0V to 3.4V

Maximum current: standard 5mm LEDs are rated for a strict continuous maximum of **20mA**. Exceeding this causes thermal runaway and permanent failure.



5.5 The Current Limiting Resistor — Critical Calculation

Because a forward-biased LED has almost no internal resistance, connecting it directly to a 5V Arduino pin would draw excessive current and destroy both the LED and potentially the Arduino pin. A series **current-limiting resistor** is mandatory.

Formula: $R = \frac{V_s - V_f}{I_f}$

Where:

- V_s = source voltage (5.0V on Arduino)
- V_f = LED forward voltage drop (e.g., 2.0V)
- I_f = desired forward current (e.g., 0.02A / 20mA)

Worked example:

1. Voltage across the resistor: $V_R = 5.0V - 2.0V = 3.0V$
2. Resistance: $R = 3.0V / 0.02A = \mathbf{150\ \Omega}$ (safe minimum)

In practice, **220 Ω** is the standard chosen value — it limits current to about 13.6mA, which is comfortably below the 20mA maximum, extending LED life and adding a further safety margin.

5.6 Practical Applications

- **Visual status indicators:** continuous ON = power confirmed; slow flashing = active sensor telemetry; fast flashing = low battery/error alert.
 - **RGB LEDs:** mix red, green, and blue elements in one housing; using PWM to vary intensity of each channel allows millions of color combinations, useful for representing proximity or temperature states.
 - **Custom display panels:** multiplexed matrices and 7-segment displays for numeric/error codes; grouped LEDs can simulate systems like traffic lights.
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Topic 6: Pulse Width Modulation (PWM)



6.1 Core Definition and the Digital Voltage Limitation

Microcontrollers like the Arduino are fundamentally **digital** — an output pin can only be strictly HIGH (5V) or strictly LOW (0V); it cannot naturally output an intermediate voltage like 2.5V. **Pulse Width Modulation (PWM)** is the technique used to *simulate* a variable analog voltage using this purely binary output.

6.2 How PWM Works

PWM rapidly switches a digital pin HIGH and LOW at high frequency. Because the physical load (an LED, a motor) cannot react instantaneously, it effectively "averages" this rapid switching into a smooth, simulated intermediate voltage.

Efficiency advantage: Traditional analog voltage regulation (like variable resistors) wastes excess energy as heat. PWM is far more efficient because the switching transistor is always in one of two states — fully ON (near-zero resistance, minimal heat loss) or fully OFF (zero current, zero heat loss) — never in a lossy in-between state.

6.3 Duty Cycle

The **Duty Cycle** is the percentage of time a signal is HIGH during one full cycle. Adjusting the duty cycle directly scales the average output voltage:

$$V_{avg} = V_{cc} \times \left(\frac{\text{Duty Cycle}}{100} \right)$$

With $V_{cc} = 5.0V$ (standard Arduino HIGH level):

- 25% duty cycle → 1.25V average
- 50% duty cycle → 2.5V average
- 75% duty cycle → 3.75V average

Arduino PWM signals typically switch at **490Hz to 980Hz** — fast enough (490–980 times per second) that the output appears smooth and flicker-free to motors, LEDs, and human perception alike.

6.4 Implementing PWM in Arduino



Not every digital pin supports PWM — only pins with dedicated hardware timers, marked with a **tilde (~)** symbol, do so. On the Arduino Uno, these are **pins 3, 5, 6, 9, 10, and 11**.

The function used is:

```
analogWrite(pin, value);
```

Where **value** is an 8-bit number from 0–255:

- **0** → 0% duty cycle (constant 0V)
- **127/128** → ~50% duty cycle (~2.5V average)
- **255** → 100% duty cycle (constant 5V)

Importantly, once **analogWrite()** is called, the microcontroller's independent hardware timer continues generating the PWM signal in the background — the rest of your code keeps executing without being blocked by the switching process.

6.5 Practical Applications

- **LED brightness fading:** varying duty cycle changes the average current through the LED; the switching is fast enough that human "persistence of vision" perceives it as smooth, flicker-free brightness change rather than blinking.
- **DC motor speed control:** PWM connected through a motor driver smoothly varies average motor voltage, avoiding the severe power loss (heat waste) that occurs when using variable series resistors instead.
- **Servo angle positioning:** Note the important distinction here — servos do **not** use PWM for power delivery, but for *communication*. The width of a periodic 50Hz pulse (typically 1ms–2ms) maps directly to a specific rotation angle (e.g., 1ms = 0°, 2ms = 180°).

Topic 7: Real-World Arduino Projects (Synthesis Exercises)



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These beginner circuits demonstrate how all the module's concepts interact together:

1. **LED Dimmer:** Reads a variable analog voltage from a potentiometer (0V–5V, `analogRead()` range 0–1023), uses the `map()` function to scale that value into the 0–255 PWM range, then outputs a PWM signal via `analogWrite()` to smoothly dim an LED.
2. **Flyback Motor Driver:** Uses a high-current transistor or MOSFET switch to control a heavy DC motor from a low-current Arduino pin. Requires a parallel flyback diode (Topic 4) to safely redirect the inductive voltage spike (Topic 2) generated when the motor is switched off, protecting the transistor and Arduino.
3. **Traffic Light Controller:** Replicates a city traffic signal using Red, Yellow, and Green LEDs, each wired through 220 Ω current-limiting resistors (Topic 5), sequenced through a structured state-machine loop in code.

Module Summary: Integrated Component Synthesis

Designing reliable robotic hardware requires understanding how these five component families work together as a system:

- **Power Stabilization (Capacitors):** act as reservoirs that smooth voltage ripples and decouple sensor lines from high-frequency power spikes.
- **Actuation & Magnetics (Inductors):** embedded in motor and relay coils to generate the electromagnetic fields needed for motion — but they generate dangerous voltage spikes when deactivated.
- **Active System Protection (Diodes):** act as circuit bodyguards, safely shunting inductive motor surges away from the Arduino and blocking reverse-polarity battery damage.
- **Visual Telemetry (LEDs):** provide efficient, low-power visual indicators, safely protected by calculated current-limiting resistors.
- **Emulated Actuation (PWM):** bridges digital code and analog-style actuation, enabling precise control of motor speed and LED brightness through purely digital switching.



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Together, these five topics form a complete conceptual toolkit for safely interfacing an Arduino's digital logic with the physical, analog world of robotics hardware.